

Assignment 2 Summary

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results

```
## # A tibble: 6 × 7
##   approach  train_n epochs val_loss val_acc test_loss test_acc
##   <chr>      <dbl>  <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 scratch    1000    12   0.587   0.693   0.587   0.701
## 2 scratch    1500    12   0.555   0.720   0.574   0.682
## 3 scratch    2000    12   0.585   0.695   0.581   0.716
## 4 pretrained 1000     6   0.279   0.881   0.252   0.894
## 5 pretrained  500     6   0.298   0.868   0.294   0.868
## 6 pretrained 1500     6   0.230   0.904   0.251   0.900
```

1. Within my knitted file, I trained with 1,000 images, a 500 validation sample, and 500 test images. The test accuracy of the scratch convolutional neural network received an accuracy rating of 70.1%. I also attached a screenshot from the knitted file above for reference. In addition, I used a couple regularization techniques to reduce overfitting, so I believe this helped test accuracy overall.
2. When I increased the training sample size to 1,500 images while still keeping validation at 500 and test at 500, the accuracy slightly went down to 68.2%. When I went up once more to 2,000 images, accuracy went up slightly so there does not seem to be a direct correlation. Increasing data does not mean accuracy will be improved due to some impact of randomness within the training. The 2% difference is within threshold and seems normal.
3. Step 2 I will keep at 1,500 images, so step 3 will be the 2,000 image dataset. I left validation at 500 and test at 500. We are now seeing 71.6% accuracy, which is an improvement. This is suggesting that scratch models may benefit by using larger datasets.
4. Test accuracy improved when increased training images with the pretrained convolutional network.
500 training images was 86.8% accuracy
1,000 training images was 89.4% accuracy
1,500 training images was 90% accuracy.
Using those three models, 1,500 was the most accurate. There was a smaller jump in accuracy between 1,000 and 1,500, so it looks like accuracy may not get much better by increasing the training images amount.

The results demonstrated that pretrained networks will outperform the scratch models, having achieved a much higher accuracy. The scratch model received a max accuracy of 71.6% with 2,000 images and the pretrained was able to get to 90% with just 1,500 images. This shows that pretrained convolutional networks are more effective when using smaller datasets.